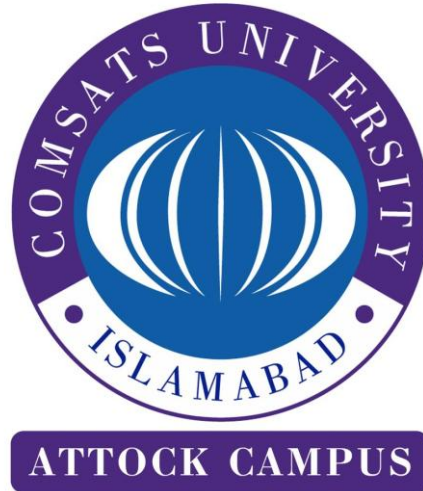


Data Structures (Lab)

Lab 2 week 03



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Lab Task 01:

```
#include <iostream>
```

```
using namespace std;
```

```
struct Node
```

```
{
```

```
    int data;
```

```
    Node* next;
```

```
};
```

```
// Insert node at end
```

```
void insert(Node*& first, int value)
```

```
{
```

```
    Node* newNode = new Node;
```

```
    newNode->data = value;
```

```
    newNode->next = NULL;
```

```
    if (first == NULL)
```

```
    {
```

```
        first = newNode;
```

```
    }
```

```
    else
```

```
    {
```

```
        Node* p = first;
```

```
    while (p->next != NULL)
    {
        p = p->next;
    }

    p->next = newNode;
}
}
```

// Display normally

```
void display(Node* first)
```

```
{
    Node* p = first;

    while (p != NULL)
    {
        cout << p->data << " ";
        p = p->next;
    }
}
```

// Reverse using loop

```
void reverseLoop(Node* first)
```

```
{
    int arr[100];
```

```
int i = 0;
```

```
Node* p = first;
```

```
while (p != NULL)
```

```
{
```

```
    arr[i] = p->data;
```

```
    i++;
```

```
    p = p->next;
```

```
}
```

```
for (i = i - 1; i >= 0; i--)
```

```
{
```

```
    cout << arr[i] << " ";
```

```
}
```

```
}
```

```
// Reverse using recursion
```

```
void reverseRecursive(Node* p)
```

```
{
```

```
    if (p == NULL)
```

```
        return;
```

```
    reverseRecursive(p->next);
```

```
        cout << p->data << " ";
    }

int main()
{
    Node* first = NULL;

    insert(first, 10);
    insert(first, 20);
    insert(first, 30);
    insert(first, 40);
    insert(first, 50);

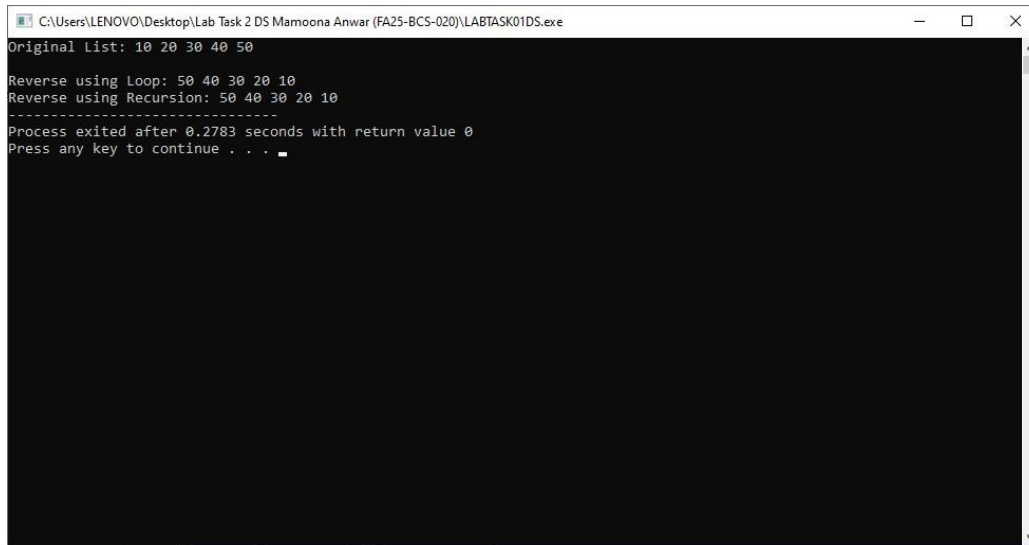
    cout << "Original List: ";
    display(first);

    cout << "\n\nReverse using Loop: ";
    reverseLoop(first);

    cout << "\nReverse using Recursion: ";
    reverseRecursive(first);

    return 0;
}
```

Output:



```
C:\Users\LENOVO\Desktop\Lab Task 2 DS Mammoona Anwar (FA25-BCS-020)\LABTASK01DS.exe
Original List: 10 20 30 40 50
Reverse using Loop: 50 40 30 20 10
Reverse using Recursion: 50 40 30 20 10
-----
Process exited after 0.2783 seconds with return value 0
Press any key to continue . . .
```

Lab Task 02:

using namespace std;

struct Node

```
{
    int data;
    Node* next;
};
```

// Insert at end

void insert(Node*& first, int value)

```
{
    Node* newNode = new Node;
    newNode->data = value;
```

```
newNode->next = NULL;
```

```
if (first == NULL)
```

```
{
```

```
    first = newNode;
```

```
}
```

```
else
```

```
{
```

```
    Node* p = first;
```

```
    while (p->next != NULL)
```

```
    {
```

```
        p = p->next;
```

```
    }
```

```
    p->next = newNode;
```

```
}
```

```
}
```

```
// Merge two lists into third list
```

```
void mergeLists(Node* first1, Node* first2, Node*& first3)
```

```
{
```

```
    Node* p = first1;
```

```
    while (p != NULL)
```

```
{  
    insert(first3, p->data);  
    p = p->next;  
}
```

```
p = first2;
```

```
while (p != NULL)  
{  
    insert(first3, p->data);  
    p = p->next;  
}  
}
```

```
// Display list
```

```
void display(Node* first)  
{  
    Node* p = first;  
  
    while (p != NULL)  
    {  
        cout << p->data << " ";  
        p = p->next;  
    }  
}
```

```
int main()
{
    Node* first1 = NULL;
    Node* first2 = NULL;
    Node* first3 = NULL;

    // First list
    insert(first1, 10);
    insert(first1, 20);
    insert(first1, 30);

    // Second list
    insert(first2, 40);
    insert(first2, 50);
    insert(first2, 60);

    cout << "First List: ";
    display(first1);

    cout << "\nSecond List: ";
    display(first2);

    // Merge
    mergeLists(first1, first2, first3);
```

```

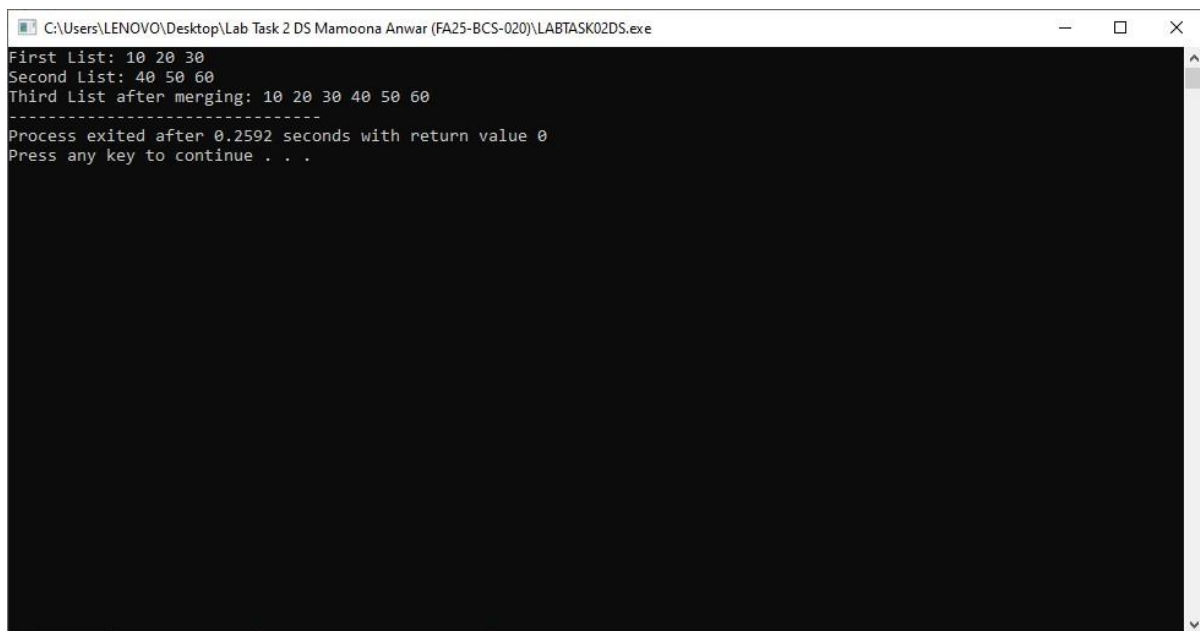
    cout << "\nThird List after merging: ";

    display(first3);

    return 0;
}

```

Output:



```

C:\Users\LENOVO\Desktop\Lab Task 2 DS Mamoon Anwar (FA25-BCS-020)\LABTASK02DS.exe
First List: 10 20 30
Second List: 40 50 60
Third List after merging: 10 20 30 40 50 60
-----
Process exited after 0.2592 seconds with return value 0
Press any key to continue . . .

```

Lab Task 03:

```

#include <iostream>

using namespace std;

struct Node
{
    int data;

    Node* next;
}

```

```
};
```

```
// Insert at end
```

```
void insert(Node*& first, int value)
```

```
{
```

```
    Node* newNode = new Node;
```

```
    newNode->data = value;
```

```
    newNode->next = NULL;
```

```
    if (first == NULL)
```

```
    {
```

```
        first = newNode;
```

```
    }
```

```
    else
```

```
    {
```

```
        Node* p = first;
```

```
        while (p->next != NULL)
```

```
        {
```

```
            p = p->next;
```

```
        }
```

```
        p->next = newNode;
```

```
    }
```

```
}
```

```

// Find multiple occurrences

void findOccurrences(Node* first, int value)
{
    Node* p = first;

    int position = 1;

    int count = 0;

    while (p != NULL)
    {
        if (p->data == value)
        {
            cout << "Found at position: " << position << endl;

            count++;
        }

        p = p->next;

        position++;
    }

    if (count == 0)
    {
        cout << "Value not found." << endl;
    }

    else

```

```
{  
    cout << "Total occurrences: " << count << endl;  
}  
}
```

// Display list

```
void display(Node* first)
```

```
{  
    Node* p = first;  
  
    while (p != NULL)  
    {  
        cout << p->data << " ";  
        p = p->next;  
    }  
}
```

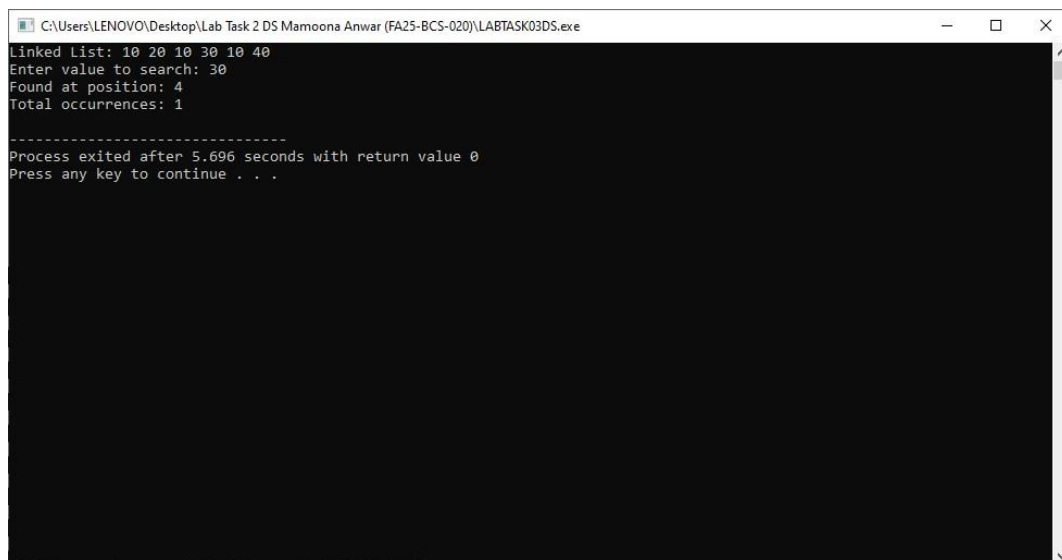
```
int main()
```

```
{  
    Node* first = NULL;  
  
    insert(first, 10);  
    insert(first, 20);  
    insert(first, 10);  
    insert(first, 30);  
}
```

```
insert(first, 10);  
insert(first, 40);  
  
cout << "Linked List: ";  
display(first);  
  
int value;  
  
cout << "\nEnter value to search: ";  
cin >> value;  
  
findOccurrences(first, value);  
  
return 0;  
}
```

Output:

S



```
C:\Users\LENOVO\Desktop\Lab Task 2 DS Mamoon Anwar (FA25-BCS-020)\LABTASK03DS.exe  
Linked List: 10 20 10 30 10 40  
Enter value to search: 30  
Found at position: 4  
Total occurrences: 1  
-----  
Process exited after 5.696 seconds with return value 0  
Press any key to continue . . .
```